Topic 1 Review · ANSWER KEY

Algebra I · Module 3, Topic 1: Linear Equations and Inequalities · Tuesday, Nov 17 · Test Wed Nov 18

Directions: This review has the same kinds of questions as the Topic 1 test, in the same order. Show your work. For multiple choice, choose one answer. Finish Part 1 (just like the test) first; Part 2 is extra practice.

Part 1 · Just like the test (Questions 1–11)

1. Jayden has $40 in his bank account. He deposits $15 each week to save for new headphones. Which equation can be used to determine the amount that Jayden has saved, y, after x weeks?

A. $y = 15 + 40x$

B. $y = 15x + 40$

C. $y = -15x + 40$

D. $y = -40x + 15$

$15 per week is the rate (multiplies x); the $40 he starts with is the constant.

2. What value of k makes the equation $5 - 4(k + 3) = 9 - 2(2k + 6)$ true?

F. 1

G. 2

H. 4

J. No solution

Left: 5 − 4k − 12 = −4k − 7. Right: 9 − 4k − 12 = −4k − 3. Same k-term, different constants → no solution.

3. What is the solution set for $3(4 - t) > -2t + 15$?

A. $t > -3$

B. $t < 3$

C. $t \geq 3$

D. $t < -3$

12 − 3t > −2t + 15 → −t > 3 → divide by −1 and flip: t < −3.

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4. Solve the equation shown.

$0.3(4x - 6) = -1.8x + 4.2$

Answer: ***x = 2***

1.2x − 1.8 = −1.8x + 4.2 → 3x = 6 → x = 2. Check: 0.3(2) = 0.6 and −3.6 + 4.2 = 0.6 ✓

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5. Does the equation shown below have one solution, no solution, or an infinite number of solutions? Explain your reasoning.

$\frac{1}{3}(12x + 30) = 4(x + 2) + 2$

Answer: **Infinite solutions: both sides simplify to 4x + 10**

Left: 4x + 10. Right: 4x + 8 + 2 = 4x + 10. The sides are identical, so every x works.

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6. Maya goes to a school store. She buys 4 notebooks that cost $1.75 each and some pens for $1.20 each. Write an equation for the total cost, *y*, when Maya buys *x* pens.

Answer: ***y* = 1.20*x* + 7.00**

Notebooks are a fixed $7.00 (4 × 1.75). Pens change: $1.20 per pen → y = 1.20x + 7.00.

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7. Rewrite the equation in slope-intercept form.
$2x + 5y = 15$

Answer: $y = -\frac{2}{5}x + 3$

5y = −2x + 15 → divide every term by 5: y = −2/5 x + 3.

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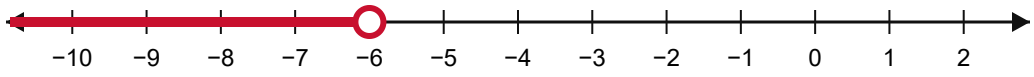
8. The formula for the volume of a rectangular prism is $V = \ell wh$. Solve the equation for h.

Answer: $h = \frac{V}{\ell w}$

h is multiplied by ℓw, so divide both sides by ℓw.

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9. Solve the inequality, and graph the solution on the number line.
$2.5m - 3 < -1.5(m + 18)$



Answer: ***m* < −6 (open circle at −6, shade left)**

2.5m − 3 < −1.5m − 27 → add 1.5m and 3: 4m < −24 → m < −6. Open circle at −6, shade left.

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10. Rewrite the equation $y - 4 = 3(x + 2)$ in slope-intercept form.

Answer: ***y* = 3*x* + 10**

y − 4 = 3x + 6 → add 4: y = 3x + 10.

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11. A triangle has a height of $(3n - 6)$ inches and a base of 10 inches. A rectangle has a length of 5 inches and a width of $(n + 4)$ inches. The area of the triangle is equal to the area of the rectangle. What is the value of n?

Answer: ***n* = 5**

½(10)(3n − 6) = 5(n + 4) → 15n − 30 = 5n + 20 → 10n = 50 → n = 5. Check: both areas are 45 in².

Part 2 · Extra practice (Questions 12–15)

12. Which equation has **no solution**?

F. $3(x - 2) = 3x + 5$

G. $3(x - 2) = 3x - 6$

H. $3(x - 2) = 2x + 5$

J. $3x - 2 = 2x - 3$

3x − 6 = 3x + 5: same x-term, different constants. (G is infinite; H and J have one solution.)

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13. What is the solution set for $-4x + 7 \geq 19$?

A. $x \geq -3$

B. $x \leq 3$

C. $x \leq -3$

D. $x \geq -6.5$

−4x ≥ 12 → divide by −4 and flip: x ≤ −3.

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14. The perimeter of a rectangle is $P = 2\ell + 2w$. Solve for w.

Answer: $w = \frac{P - 2\ell}{2}$

Subtract 2ℓ from both sides, then divide by 2: $w = (P - 2\ell)/2$ (or $w = P/2 - \ell$).

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15. A gym charges a \$25 sign-up fee plus \$8 per class. Lena can spend at most \$90. Write and solve an inequality for the number of classes, c, she can take.

Answer: $25 + 8c \le 90 \rightarrow c \le 8.125$, so at most 8 classes

$25 + 8c \le 90 \rightarrow 8c \le 65 \rightarrow c \le 8.125$. Classes are whole numbers, so 8.
